Homework 5 Solution

Instructor solution notebook. All problems are answered in writing.

Problem 1: Regression $y = w^2 x + wx$

Let $a(w) = w^2 + w$, so the model is $y = a(w)\, x$. The least-squares loss is

$$L(w) = \sum_{i=1}^{n} (y_i - \left(w^2 + w\right) x_i)^2.$$

Stationarity condition

$$\frac{dL}{dw} = -2(2w + 1) \sum_i x_i (y_i - \left(w^2 + w\right) x_i) = 0.$$

Define $\widehat{a} = \dfrac{\sum_i x_i y_i}{\sum_i x_i^2}$ (the ordinary least-squares slope through the origin). Then

$$\frac{dL}{dw} = -2(2w + 1)(\sum_i x_i^2)(\widehat{a} - w^2 - w),$$

so the stationarity condition is the cubic $2w^3 + 3w^2 + \left(1 - 2\widehat{a}\right) w - \widehat{a} = 0$, which factors as

$$(2w + 1)\left(w^2 + w - \widehat{a}\right) = 0.$$

Solutions

- $w = -\frac{1}{2}$ (always a real root). Here $a(w) = -\frac{1}{4}$ is the minimum of $w^2 + w$. It is the loss minimizer only when $\widehat{a} < -\frac{1}{4}$ (see below); otherwise it is a local maximum of L.
- $w^2 + w - \widehat{a} = 0$, so $w = \dfrac{-1 \pm \sqrt{1 + 4\widehat{a}}}{2}$. These are real iff $\widehat{a} \geq -\frac{1}{4}$.

Which root is the least-squares solution?

Since $L = \sum_i \left(y_i - a x_i\right)^2$ is a convex quadratic in $a = w^2 + w$, and $a(w)$ takes every value in $[-\frac{1}{4}, \infty)$:

- **If** $\widehat{a} \geq -\frac{1}{4}$ (with $\sum x_i^2 > 0$): $a = \widehat{a}$ is attainable, and the global minimizers are the two roots

  $$w^\star = \frac{-1 \pm \sqrt{1 + 4\widehat{a}}}{2}, \qquad \widehat{a} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}$$

  Both give the same loss, because w and $-1 - w$ give the same $w^2 + w$. At $\widehat{a} = -\frac{1}{4}$ they coincide at $w = -\frac{1}{2}$.
- **If** $\widehat{a} < -\frac{1}{4}$: the quadratic has no real roots, the only real stationary point is $w = -\frac{1}{2}$, and it is the global minimizer (the closest attainable value $a = -\frac{1}{4}$).

Problem 2: Can the MLE be found by linear regression?

With iid Gaussian noise, the MLE of θ minimizes the sum of squared errors. It is found by *linear* regression exactly when the mean function is linear in the parameters θ (it may be nonlinear in x).

Part (a): $y = \theta_1 x \sin(x) + \epsilon$

True. The mean is $\theta_1 \phi(x)$ with fixed feature $\phi(x) = x \sin x$, linear in θ_1. Regress y on $\phi(x)$: $\widehat{\theta_1} = \sum \phi_i y_i / \sum \phi_i^2$.

Part (b): $y = x^{\theta_1} e^{\theta_2} + \epsilon$

False. The mean is nonlinear in θ_1 (it appears in an exponent), and e^{θ_2} is a nonlinear function of θ_2. Taking logs would linearize the *noiseless* model, but the noise is additive, so $\log y \neq \theta_1 \log x + \theta_2 + \epsilon$ and the log-regression does not give the MLE (nonlinear least squares is required).

Part (c): $y = \log\left(x^{\theta_1} e^{\theta_2}\right) + \epsilon$

True (for $x > 0$). $\log\left(x^{\theta_1} e^{\theta_2}\right) = \theta_1 \log x + \theta_2$, which is linear in $\theta = (\theta_1, \theta_2)$ with features $[\log x,\ 1]$. Ordinary linear regression gives the MLE.

Problem 3: MLE for $p(y;\theta) = \frac{y}{\theta} \exp\left(-\frac{y^2}{2\theta^2}\right)$

Assume $y_i > 0$ and $\theta > 0$. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} \frac{y_i}{\theta} \exp\left(-\frac{y_i^2}{2\theta^2}\right),$$

so the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \left[\log y_i - \log\theta - \frac{y_i^2}{2\theta^2}\right] = \sum_i \log y_i - n \log\theta - \frac{1}{2\theta^2} \sum_i y_i^2.$$

Setting the derivative to zero,

$$\frac{d\ell}{d\theta} = -\frac{n}{\theta} + \frac{1}{\theta^3} \sum_i y_i^2 = 0 \Rightarrow \theta^2 = \frac{1}{n} \sum_i y_i^2.$$

The second derivative at this point is $\frac{n}{\theta^2} - \frac{3}{\theta^4} \sum y_i^2 = \frac{n}{\theta^2} - \frac{3n}{\theta^2} = -\frac{2n}{\theta^2} < 0$, so it is a maximum:

$$\theta_{\text{MLE}} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} y_i^2}$$

Remark. As written, $\int_0^\infty \frac{x}{\theta} e^{-x^2/2\theta^2} dx = \theta \neq 1$, so this is not a normalized density; the standard Rayleigh density is $\frac{x}{\theta^2} e^{-x^2/2\theta^2}$, whose MLE is $\theta^2 = \frac{1}{2n} \sum y_i^2$. The problem asks for the MLE of the likelihood as given, which is the boxed result.